

LESSON 115 (1.0 GROUND and 1.5 DUAL)

Lesson Objective

The purpose of this lesson is to introduce the student to emergency procedures, steep turns, and non-towered airport operations. During this lesson the student will perform the procedure for an engine failure in flight with emphasis on the emergency approach and landing pattern.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Warrior Maneuver's Manual.
 - Non-towered operations.
 - Steep turns.
 - Emergency checklists.
 - V speed and Emergency quiz review.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Non-Towered Airport Flight Operations. ([AC 90-66](#))
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 08 - Airport Traffic Patterns.
 - Chapter 10 - Performance Maneuvers.

Lesson Content (Ground)

Non-Towered Operations

- Entry:
 - Determine the weather at the field.
 - Determine how to enter the pattern.
 - Enter 45° to the downwind leg.
 - Overfly the field at 1500 ft.
 - Radio calls must be made within 10 NM of the airport.
- Traffic Pattern:
 - Radio calls must be made on every leg.
 - The pattern should be flown at the appropriate traffic pattern altitude.
 - Cross-wind should be turned at 300' below traffic pattern altitude
- Exiting the Traffic Pattern:
 - When departing the traffic pattern, continue straight out, or exit with a 45° turn in the

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direction of the traffic pattern beyond the departure end of the runway. Departures must climb to the traffic pattern altitude before turning.

Radio Communications in Non-Controlled Airspace

- Although radio communication in non-controlled airspace are not legally required, it is highly recommended and FITA requires it!
 - Collision avoidance.
- Radio calls must be made within 10 NM, and during every leg of the pattern.
 - Distance can be obtained by going to the nearest page on the G430.
- All radio calls in a non-controlled airspace should be made using the full tail number.
 - Ex: "Valkaria traffic, N602FT is 10 NM north of the airport and will be entering 45 degrees to the left downwind for runway 10, Valkaria traffic."

Engine Failure in Flight (Emphasis on Pattern)

- Order of operations
 - A - Airspeed
 - Airspeed comes first and is critical to maintain throughout the maneuver because it guarantees you the most possible time in the air to make a safe landing.
 - Review why we use V_{Glide} (73) and not another airspeed.
 - Total drag curve.
 - Show where 73 is and why going at 63 or 83 (for example) would be worse.
 - B - Best field
 - After you can keep yourself in the air for as long as possible, the next most important consideration is where to put the airplane when that time runs out.
 - C - Checklists
 - After you know you can put the airplane safely on the ground somewhere, the next step to increase the likelihood of a safe outcome is to try and restart the engine. The emergency checklists should be memorized in a flow pattern. If altitude permits, it is also recommended to verify no line items were missed with the physical checklist.
 - D - Declare
 - Aviate, Navigate, Communicate. If safe to do so, declare an emergency on 121.5 and squawk 7700.
 - "mayday mayday mayday (tail number) (location) (emergency) (souls on board) mayday mayday mayday".
 - You can press and hold the frequency toggle soft button to automatically put 121.5 in the active frequency box in COM 1.

Emergency Approach and Landing

- This is performed to simulate a situation where power is lost in flight.
- Ensure that the student is familiar with the procedures and has the checklists memorized!
- Quiz the student on the procedures and address any questions they might have.

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- This is performed to simulate a situation in which the airplane needs to descend in the most expeditious means possible.
- Smoke in the cockpit, electrical fire, hypoxic passenger, depressurization, general emergency situations.
- Quiz the student on the procedures for the maneuver and address any questions they might have.

Steep Turns

- Steep turns consist of one or multiple 360° turns in one or both directions with a bank of 45°.
- Steep turns help a student to develop their sense of coordination and attention switching capabilities.
- More details are located in lesson content below.

Normal Traffic Pattern and Powered Approach

- Ensure that the student knows how to conduct a proper traffic pattern by quizzing them on:
 - Entry procedures.
 - Airspeeds in the pattern.
 - Flap usage in the pattern.
 - Power settings.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)**Preflight Inspection**

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.

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- IF YOU SEE SOMETHING, SAY SOMETHING!

Normal/Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.

Review Previous Exercises as Required

- Review any areas the student might need help with by utilizing the pertinent lesson plans.

Introduction to Steep Turns

- Instructor does, student says:
 - Entering the turn:
 - Ensure pre-maneuver check is complete.
 - Choose a visual reference point and bug it in the HSI.
 - Smooth level turn to 45 degrees of bank (too aggressive with the bank and back pressure can result in an accelerated stall!)
 - At 30 degrees of bank, add 100-200 RPM and 3 rolls of aft trim.
 - Maintaining the turn:
 - Focus should primarily be outside.
 - Show the student where the horizon intersects with the panel/cowling. The student should maintain that sight picture.
 - Coordination.
 - Reference the instruments.
 - Corrections:
 - If the nose falls below the desired sight picture, pitch needs to increase.
 - If the nose rises above the desired sight picture, pitch needs to decrease.
 - Power should only be used to correct for airspeed changes.
 - Rolling out:
 - Roll-out should commence at $\frac{1}{2}$ the bank angle.
 - Pitch the nose down to counteract the decrease in horizontal lift and increase in vertical lift.
 - Bring the power back to cruise power.
 - Trim for level flight.
- Student says, student does:
 - Verify correct procedure.

Engine Failure in Flight (Emphasis on Pattern)

- Student says, student does.
 - Verify correct procedure.
- Points of Emphasis.
 - Trimming for V_{glide} .

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- Wire strike avoidance.
- Choice of landing point.
- Coach the student through a proper flow while going through the checklist (left to right!).

Emergency Descent

- Student says, student does.
 - Verify correct procedure.
- Points of Emphasis.
 - Trim.
 - Area clearance.
 - Ensure the student does not exceed V_{FE} .

Normal Traffic Pattern and Powered Approach in Non-Towered Airport

- Proceed to Valkaria or Sebastian.
- Ensure the descent/arrival checklist is conducted.
- Guide the student through:
 - Obtaining the weather.
 - Figuring out which runway to use.
 - Determining the entry.
- The student should be making the radio calls.
 - Provide guidance as needed.
- Traffic patterns should be conducted by the student.

Normal/Crosswind Landing

- The student should conduct the landing.
- Provide guidance during the round and flare.
- Provide guidance on proper crosswind correction.
 - Emphasize that this must be maintained even after touchdown.

Go-around/Rejected Landing

- Have the student conduct a go-around during one of the traffic patterns.
- Ensure that the correct procedure is conducted.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.

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- Schedule the next lesson.
- Preview the next lesson → 116.
- Assign homework:
 - Review the emergency checklists.
 - Review ground reference maneuvers.
 - Give a blank V speed and Emergency quiz, inform the student that it will be tested on 118 Mock.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
3. Perform steep turns with instructor guidance.
4. Conduct emergency procedures with instructor guidance.
5. Conduct non-towered traffic patterns with instructor guidance.
6. Conduct takeoffs and landings with little instructor guidance.